

Solar Rooftop PV Techno-Commercial Case Study

5.175 kWp Grid-Connected Residential Rooftop Solar System

A self-directed mock techno-commercial case study, prepared to demonstrate a customer-requirement-to-proposal workflow

1. Executive Summary

This is a self-directed mock techno-commercial case study, built to demonstrate a complete requirement-to-proposal workflow for a rooftop solar system. It uses a simulated residential customer, public manufacturer datasheets, and publicly referenced market prices - it is not a live client engagement, and no equipment quotation has been sought from any vendor.

Customer problem: a residential customer in the Delhi-NCR region has a high monthly electricity bill (assumed Rs. 5,500 at Rs. 8/kWh) and wants to reduce it through rooftop solar, while keeping upfront cost moderate.

Proposed solution: a 5.175 kWp grid-connected system using 9 x Waaree 575 Wp N-Type bifacial modules and a Loom Solar Fusion 63 (6 kW, three-phase) inverter.

System size	5.175 kWp DC
Estimated annual generation (Year 1)	6,800 kWh
Gross project price (before subsidy)	Rs. 2,78,349
PM Surya Ghar subsidy (CFA)	Rs. 78,000
Net customer investment	Rs. 2,00,349
Net annual savings (Year 1)	Rs. 47,310
Simple payback	~4.2 years

Commercial conclusion: under the stated assumptions, the system is commercially attractive - a mid-4-year payback with government subsidy support and a 25+ year useful asset life. The case study also works through a genuine component-sizing trade-off (inverter selection) explicitly, and is transparent about which figures are assumptions versus verified facts.

2. Customer Requirement & Load Assessment

Simulated customer profile: an urban residential homeowner in the Delhi-NCR region, with independent rooftop access and a moderate-upfront-cost priority (not necessarily the maximum possible system size or the absolute lowest payback).

Item	Value	Basis
Avg. monthly electricity bill	Rs. 5,500	Assumption - simulated customer
Assumed electricity tariff	Rs. 8/kWh	Assumption - Delhi-NCR mid-slab residential
Derived monthly consumption	687.5 kWh	$= \text{Bill} \div \text{Tariff}$
Derived annual consumption	8,250 kWh	$= \text{Monthly consumption} \times 12$
Usable shadow-free roof area	550-600 sq ft	Assumption
Target offset of annual consumption	85%	Assumption - 100% would oversize vs. roof area/budget
Target annual generation	7,012.5 kWh	$= \text{Annual consumption} \times 85\%$

No assumptions beyond those already fixed in the finalized workbook have been introduced here.

3. Solar System Sizing

The target annual generation is converted into an indicative system capacity using standard planning inputs for the region, then rounded to a practical whole-module configuration.

Parameter	Value	Reason
Peak Sun Hours (PSH)	4.5 hrs/day	Conservative planning value for Delhi-NCR
Performance Ratio (PR)	80%	Combined module/inverter/temperature/wiring/soiling losses
Indicative required capacity	~5.34 kWp	$= \text{Target generation} \div (\text{PSH} \times 365 \times \text{PR})$
Selected module rating	575 Wp	Waaree N-Type bifacial module (see Section 4)
Number of modules	9	Closest whole-module fit to indicative requirement, within assumed roof area
Resulting DC system size	5.175 kWp	$= 9 \times 575 \text{ Wp}$
Estimated annual generation, Year 1	6,800 kWh	$= 5.175 \times 4.5 \times 365 \times 0.80$ (see Section 6)

Why 9 modules: nine 575 Wp modules (5.175 kWp) is the closest whole-module fit to the ~5.34 kWp indicative requirement, and comfortably suits the assumed 550-600 sq ft roof area. The resulting generation (6,800 kWh) is within 15% of the original 7,012.5 kWh target - an acceptable practical fit, not an exact match, since module counts are necessarily whole numbers.

4. Technical Design & Component Selection

4.1 Module - Waaree BIN-08-575 (575 Wp, N-Type TOPCon, Bifacial)

Technology	N-Type TOPCon, bifacial, dual-glass, monocrystalline
Rated power / Efficiency	575 Wp / 22.26%
Voc / Isc	52.3 V / 13.87 A
Warranty	12-yr product / 30-yr linear power output

Why this module: it offers a strong efficiency-to-cost balance and is domestically manufactured, which is required for PM Surya Ghar subsidy eligibility (Section 6). Bifacial technology can add a further generation uplift, but this case study conservatively excludes any bifacial gain from the 6,800 kWh/year headline figure, since a flush-mounted residential installation captures little rear-side reflected light in practice.

4.2 Inverter - Loom Solar Fusion 63 (6 kW, Three-Phase)

A grid-tie inverter is required to convert the array's DC output to grid-synchronized AC and to provide safety functions such as anti-islanding protection.

The 6 kW three-phase Fusion 63 provides sufficient AC capacity for the 5.175 kWp array. The resulting DC/AC ratio is approximately 0.86. A smaller single-phase configuration (~4 kW class) was also considered, since it would give a DC/AC ratio closer to the typical 1.1-1.25 EPC target; the selected configuration was retained under the assumed three-phase customer connection and to keep design headroom. In an actual project, the customer's electrical connection and the exact inverter's operating window would be confirmed on-site before finalizing the design.

Check	Result	Note
String Voc (9 modules in series)	470.7 V	Well within the inverter's 1000V max input, even allowing for cold-weather Voc increase
String Isc	13.87 A	Comfortably within the inverter's 40A max input current
DC/AC ratio	0.86	Below the typical 1.1-1.25 target - a design trade-off, not a sizing error; the array is never clipped
Max PV input power / voltage	7.8 kW / 1000 V	
MPPT trackers	2	
Max efficiency	98.5%	
Protection	Anti-islanding, over-voltage/current/temperature, DC reverse polarity, Type II SPD	

5. BOM & Techno-Commercial Costing

Costing follows a standard build-up: itemized materials and balance-of-system components, installation and engineering, a contingency allowance, and an EPC commercial margin to arrive at a customer-facing gross price.

Module and inverter costs are public manufacturer/retail reference prices; all balance-of-system, installation and engineering line items are engineering estimates typical for a residential job of this size - none of these figures are vendor quotations.

Item	Basis	Amount (Rs.)
Waaree 575 Wp module (9 nos)	Reference price	1,01,691
Loom Solar Fusion 63 (1 no)	Estimated (Rs./kW benchmark)	42,000
Mounting, cabling, protection, earthing, monitoring (BOS)	Estimated	64,000
Installation, transport, engineering, testing	Estimated	29,000
Subtotal	-	2,36,691
Contingency (5%)	Estimated	11,835
Total Direct Project Cost	-	2,48,526
EPC overhead & commercial margin (12%)	Rs. 29,823	
Gross Project Price (before subsidy)	Rs. 2,78,349	
PM Surya Ghar CFA subsidy	Rs. 78,000	
Net Customer Investment	Rs. 2,00,349	

6. PM Surya Ghar Subsidy & Financial Analysis

6.1 Subsidy Structure (official MNRE policy, not an assumption)

Slab	Rate
1st kW	Rs. 30,000/kW
2nd kW	Rs. 30,000/kW
3rd kW	Rs. 18,000/kW
Cap for systems ≥ 3 kW (residential)	Rs. 78,000 flat

This 5.175 kWp system exceeds 3 kW, so the flat Rs. 78,000 residential cap applies rather than a per-kW slab calculation. Eligibility also requires domestically manufactured modules - Waaree is a domestic manufacturer, so this system qualifies.

6.2 Annual Savings Calculation

Generated units are not saved 1:1 at the retail tariff. Self-consumed units offset the bill at the full retail rate; exported units are credited at the (lower) net-metering rate.

Step	Value
Estimated annual generation, Year 1	6,800 kWh
Self-consumption share (assumption)	55%
Self-consumed units	3,740 kWh/yr
Exported units	3,060 kWh/yr
Savings from self-consumption (@ Rs. 8/kWh)	Rs. 29,920/yr
Credit from grid export (@ Rs. 6.5/kWh)	Rs. 19,890/yr
O&M cost	(Rs. 2,500/yr)
Net Annual Savings, Year 1	Rs. 47,310

6.3 Payback & Long-Term View

Net customer investment	Rs. 2,00,349
Simple payback (Net investment $\div$ Net annual savings)	4.23 years
Approx. 25-year cumulative net savings	~Rs. 10.6 lakh
Approx. 25-year return on net investment	~530%

The 25-year figure is a simplified cumulative view (Year-1 savings projected forward with an approximate degradation allowance and one inverter replacement at Year 12) - it is explicitly NOT a discounted cash-flow (DCF) analysis. Simple payback is used as the headline metric because it is what a residential customer typically asks about directly.

7. Sensitivity Analysis

The business question this answers is: *what happens to the commercial case if our assumptions are wrong?* Three variables were each moved independently by a realistic margin.

Variable	Lower / Low	Base	Higher / High
Electricity tariff (Rs.6.5 / Rs.8 / Rs.9.5 per kWh)	4.80 yrs	4.23 yrs	3.79 yrs
Annual generation (-10% / base / +10%)	4.73 yrs	4.23 yrs	3.83 yrs
Project cost (-10% / base / +10%)	3.81 yrs	4.23 yrs	4.66 yrs

Payback stays within a 3.8-4.8 year band across all three variables moving independently by 10% - the base case (~4.2 years) is not fragile to any single assumption moving moderately in an unfavourable direction.

8. Commercial Recommendation

Recommendation: under the stated assumptions, the proposed 5.175 kWp system is commercially attractive - it offers a mid-4-year payback, direct government subsidy support (reducing upfront cost by ~28%), and a 25+ year asset life with strong long-run returns once payback is reached. The selected components balance efficiency, warranty quality, and cost for a customer prioritizing moderate upfront investment over the absolute lowest payback.

Key risks acknowledged: actual generation depends on real irradiation, shading and orientation and could vary from this model's planning assumptions; the inverter has a shorter expected life than the module warranty and will likely need one replacement; net-metering policy and export credit rates can change at the DISCOM/state level; and the selected inverter configuration assumes a three-phase customer connection that has not been physically verified.

Before any real quotation could be issued, the following would need to be validated:

To be validated	Why
Site survey (shading, orientation, structural condition)	Directly affects real achievable generation and mounting design
Actual electricity consumption and tariff slab	This model uses an assumed bill and flat tariff, not a real bill
Electrical phase/connection type	The selected inverter assumes three-phase; this must be confirmed on-site
Exact equipment datasheets and supplier quotations	BOS/installation costs here are estimates, not vendor quotes
Net-metering application requirements (DISCOM-specific)	Process and export credit terms vary by state/DISCOM
Installation scope and warranty terms	Final contract terms would need to be negotiated with the EPC contractor
Subsidy eligibility confirmation	PM Surya Ghar terms/eligibility should be reconfirmed at the time of application

9. Relevance to Technical Sales / Business Development

This case study was built to practice the workflow and thinking relevant to a technical sales, business development and product management role - it does not claim to be equivalent to professional industry experience.

Role-relevant requirement	Demonstrated through this project
Customer requirement understanding	Translated a stated bill/budget/roof constraint into a sized technical solution (Section 2-3)
Technical solution mapping	Matched system capacity and component specs to the customer's stated priorities (Section 3-4)
Component selection & compatibility	Verified string voltage/current compatibility and identified a genuine sizing trade-off (Section 4)
Market/price benchmarking	Anchored module and inverter costs to public reference prices (Section 5)
BOM and costing	Built a itemized, transparent cost structure distinguishing reference prices from estimates (Section 5)
Business case & pricing	Applied EPC margin and government subsidy to reach a customer-facing price (Section 5-6)
Customer-facing recommendation	Balanced recommendation with risks and next steps, not a one-sided pitch (Section 8)
Sensitivity / handling uncertainty	Quantified how payback shifts under tariff/generation/cost variation (Section 7)
Technical-commercial trade-offs	Explicitly reasoned through and documented the inverter sizing trade-off (Section 4)